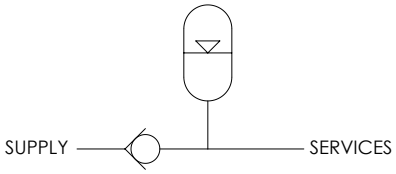
Introduction

BiS Valves offer an extensive range of check and shuttle valve products, in cartridge, manifold mounted and ported installation formats.

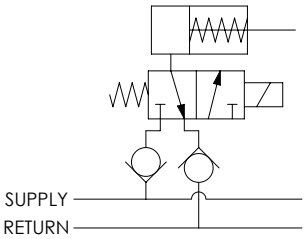
Check Valves

Check valves, sometimes known as non-return valves, NRV's or one-way valves allow flow in one direction, and block flow from returning. Nearly all hydraulic and pneumatic circuits contain check valves as a critical part of the functionality. Typical applications include prevention of back flow into pumps, accumulator unloading systems, and the prevention of undesirable actuator movements when directional valves are operated.



Check valve prevents filled accumulator discharging when supply is unloaded.

Typical Uses

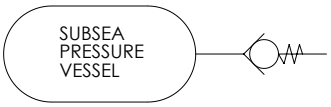


Check valves prevent unintended actuator movements during transient pressure spikes or drops in supply and return lines.

Venting Check Valves

Venting check valves have a single threaded connection and are often used to allow fluid out of a system. They are used topside in venting applications and find many uses in subsea pressure compensation applications.

Typical Uses

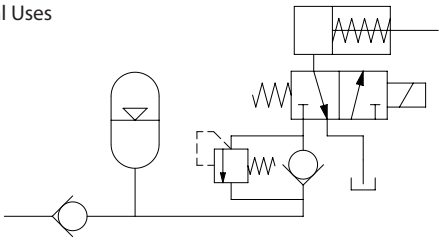


Venting check valve maintains a small positive pressure differential inside the vessel during a raising operation.

Yield Valves

Yield valves are essentially check valves with an integral reverse relief function, intended to provide thermal relief for small trapped volumes of fluid. A typical example would be the supply line to a directional control valve, where increased pressure due to thermal expansion could overpressurise the directional valve.

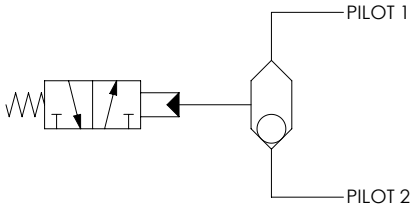
Typical Uses



Where pressure is trapped between the check valve and the directional valve or actuator, the integral relief function in a yeld valve can limit pressure increase due to thermal expansion of the trapped volume.

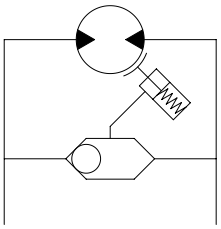
Shuttle Valves

Shuttle valves are used to automatically select the higher of two input pressures. They are commonly used to combine dual redundant supply lines, emergency hydraulic power sources and for brake release applications when driving hydraulic motors.



Two pilot supplies are combined via a shuttle valve such that if one pilot supply fails, the other pilot supply can still operate the directional valve.

Typical Uses



Shuttle valve delivers a brake release signal to bi-directional motors.

Check and Shuttle
Overview

Selecting the Correct Product

Selection guide for check valves

	Page	Suitable for liquids	Suitable for gases	1/4" – 3/8" BSP / NPT ports	1/2" – 1" BSP/NPT ports	> 1" BSP/ NPT ports	Medium pressure ports	Cartridge installation	Manifold mount installation	Venting check valve	Thermal relief function
C25	5										
C37	6										
C50	7										
C75	8										
C100	9										
C125	10										
C150	11										
NR25	12										
NR25 (20K)	13										
NR37	14										
NR50	15										
NR50 (20K)	16										
NR50M	17										
NR75	18										
NR100	19										
NR200	20										
CC25	21										
CC37	22										
CC50	23										
CC75	24										
CC100	25										
CV25	26										
YV25	27										

Important – This is a guide to selecting products that may be suitable. The valve data sheet and application notes must be thoroughly reviewed prior to final selection. Please contact us if the options you require are not shown

Selection guide for shuttle valves

	Page	Suitable for liquids	Suitable for gases	1/4" – 3/8" ports	1/2" – 1" ports	Cartridge installation
SV25	28					
SV25C	29					
SV50	30					

Important – This is a guide to selecting products that may be suitable. The valve data sheet and application notes must be thoroughly reviewed prior to final selection. Please contact us if the options you require are not shown